VARUN VINOD SINGH

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Analytics Engineer | Power BI · Microsoft Fabric · DAX · Semantic Models · SQL · Dimensional Modeling · dbt

SUMMARY

Analytics engineer with nearly 3 years building semantic models, dimensional data models, and curated datasets in Microsoft Fabric, Power BI, SQL, and dbt. Built a **7-table** Fabric semantic model for **4 station teams** at Northeastern and a self-serve model for **3 clinical operations teams** at Mass General Brigham; at LTM, cut data incidents **45%** with validation controls. Experienced preparing AI-ready datasets for retrieval use cases with Databricks and Unity Catalog.

EDUCATION

Northeastern University, M.S. Information Systems — **Sept 2024 – Dec 2026**

Mumbai University, B.E. Information Technology — **May 2022**

WORK EXPERIENCE

Data & Automation Analyst, Study Space Operations | Northeastern University — **Jan 2026 – Present**

Boston, MA · Microsoft Fabric, Power BI, DAX, SQL, PySpark, Delta Lake, Fabric Data Pipeline, Dataverse

- Designed a Power BI semantic model on a **7-table** star schema in Microsoft Fabric with **4 station teams**, so every report draws on one consistent set of measures and definitions.
- Built hourly Fabric pipelines curating **112 daily submissions** from **4 sources** into validated Delta Lake tables, so every semantic model refresh runs on clean, current data.

Digital Intern | Mass General Brigham — **June 2025 – Dec 2025**

Boston, MA · SQL, Python, PySpark, Microsoft Fabric, Power BI, MySQL, ServiceNow, Azure DevOps

- Translated clinical requirements into a ServiceNow and MySQL model that let **3 clinical operations teams** self-serve insights.
- Automated weekly ETL of **140,000+ metadata records** with classification and quality validation, so governance stakeholders could trust every weekly refresh.
- Anonymized PHI across **100,000+ records** while joining structured and unstructured data, producing governed, AI-ready datasets for downstream models.
- Moved the team from a Miro board to an Azure DevOps Scrum cycle, running sprints and stand-ups that gave every task a clear owner.

Data Engineer | LTM Limited — **Sep 2022 – Jul 2024**

Mumbai, India · SQL, Python, Java, Spark, Kafka, Airflow, Git, CI/CD, AWS S3, Glue Data Catalog, MySQL, MongoDB

- Cut data incidents **45% (11 to 6 monthly)** with SQL validation, type checks, and deduplication that kept bad records out of reports.
- Reduced pipeline maintenance **30% (10 to 7 hours weekly)** by standardizing schema contracts with **5 engineering teams**, freeing 3 engineering hours every week.
- Built **3 data pipelines** processing **1,500 records/minute** from 5 sources, with SLA monitoring, retries, and backfill recovery.
- Saved **5+ engineering hours per sprint** across 20 weekly deployments by automating Docker builds with AWS CI/CD on every merge.

PROJECTS

Semiconductor Yield Analytics Platform | *Databricks, Delta Lake, dbt, PySpark, Power BI* — **July 2026**

- Built a Databricks medallion lakehouse with Gold-layer yield and utilization models in dbt over **50M+** records, reported in Power BI.
- Cut root-cause investigation time **35%** with automated SPC validation checks in dbt that flag process deviations early.

SEC Filing Search Engine | *dbt, DuckDB, Dagster, Qdrant, Streamlit, Anthropic API*

- Built a retrieval-augmented generation (RAG) pipeline over SEC filings, modeling filing text with dbt on DuckDB in Dagster.
- Indexed curated filings in Qdrant and served LLM-powered search in Streamlit, so answers draw only from curated source data.

TECHNICAL SKILLS

Semantic Modeling & BI: Power BI, DAX, Microsoft Fabric semantic models, measures and KPIs, refresh strategies, Tableau

SQL & Modeling: SQL (window functions, query optimization, incremental loads), dimensional modeling, star schema, SCD Type 2, dbt

Data Platforms: Microsoft Fabric, Databricks, Delta Lake, medallion architecture, Unity Catalog, Snowflake

Governance & Quality: Data validation, dbt tests, DQX, pandera, metadata classification, lineage, PHI anonymization

AI-Ready Data & Delivery: RAG (Qdrant, pgvector), Python, PySpark, Azure DevOps, GitHub, CI/CD, Scrum